
Mixing Matters: Evidence Position Bias Across Attention and State-Space Language Models

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Abstract

Language models often answer multi-document questions less accurately when the relevant document appears in the middle of the context. We ask whether the sequence mixer changes this accuracy-versus-position curve under approximately matched pretraining data, scale, and context. Our deterministic evaluation moves one gold document through ten positions while holding the question and nine distractors fixed. We summarize start and end advantages with primacy and recency edges and use a question-level paired bootstrap with 10,000 resamples and Holm correction. On 800 exploratory questions, Pythia-2.8B has a positive primacy edge (+0.052, 95% CI [0.032, 0.072]), while two Pile-trained Mamba variants do not (−0.001 and −0.018); both paired architecture interactions exclude zero. A more tightly matched 8B Mamba-2 pair is directionally consistent: the hybrid containing 7% attention layers has a larger primacy edge than the pure model, but the paired difference is uncertain (+0.019, 95% CI [−0.006, 0.044], Holm $p = 0.144$). Scale, corpus, synthetic retrieval, prompt, attention-sink, and linear-probe analyses further bound the result. Together they establish an exploratory architecture-position interaction and motivate, but do not yet prove, an attention-based mechanism.

1 Introduction

Transformers often read multi-document contexts unevenly: accuracy peaks when the answer-bearing document is near an edge and falls in the middle (Liu et al., 2024). This behavior matters because retrieval systems cannot assume that all evidence positions are equally usable. Prior work links lost-in-the-middle behavior to U-shaped attention patterns (Hsieh et al., 2024a), but model-family comparisons still confound sequence mixing with data, tokenizer, scale, depth, and positional encoding. State-space models such as Mamba (Gu and Dao, 2023; Dao and Gu, 2024) provide a useful contrast because they compress the prefix into a recurrent state instead of using dense causal attention.

We make three contributions. First, we build an auditable ten-position evaluation with floor and ceiling anchors, preserved generations, and inference settings recorded per example. Second, we find a paired architecture-position interaction in a Pile-trained 2.8B comparison and test it with a more tightly matched 8B pure/hybrid pair. Third, we organize scale, corpus, task, prompt, sink, and probe results into mechanistic evidence with explicit uncertainty. All reported results use an exploratory split; the held-out confirmatory split remains unspent.

33 2 Related work

34 **Position bias in long contexts.** Liu et al. (2024) established the lost-in-the-middle curve in re-
35 trieval and multi-document QA. Subsequent work measures effective context length (Hsieh et al.,
36 2024b), associates the curve with attention allocation (Hsieh et al., 2024a), and proposes context-
37 use interventions (An et al., 2024; Peysakhovich and Lerer, 2023). We focus on whether the curve
38 changes across sequence mixers.

39 **Sequence mixers.** Dense self-attention (Vaswani et al., 2017) mixes token pairs under a causal
40 mask, while Mamba and Mamba-2 carry information through selective state-space recurrences (Gu
41 and Dao, 2023; Dao and Gu, 2024). Hybrid models interleave the two mixers (Lieber et al., 2024;
42 NVIDIA, 2025). Waleffe et al. (2024) release the pure and hybrid 8B Mamba-2 checkpoints used in
43 our tightest controlled comparison.

44 **Attention sinks.** Xiao et al. (2024) showed that causal transformers can concentrate attention on
45 initial tokens. We test whether this attention-sink statistic covaries with primacy, treating the result
46 as correlational because a valid sink-blocking ablation is absent.

47 3 Measurement and statistics

48 **Task and anchors.** We use the 2,655 released questions from the Lost-in-the-Middle 10-document
49 QA setting (Liu et al., 2024). Each question has one gold document and nine distractors. We build
50 ten prompts per question, moving only the gold document through positions 0 to 9 while retaining
51 the same distractors and token budget. A no-gold floor measures guessing or memorized knowledge,
52 and a gold-only ceiling measures answerability under perfect retrieval.

53 **Edges and statistics.** Let a_S , a_M , and a_E be mean accuracy at start positions $\{0, 1\}$, middle
54 positions $\{4, 5\}$, and end positions $\{8, 9\}$. We define primacy as $a_S - a_M$ and recency as $a_E - a_M$.
55 The question is the unit of analysis. Each of 10,000 bootstrap samples draws complete ten-position
56 question bundles with replacement, and model comparisons reuse the same sampled questions. We
57 report percentile 95% confidence intervals and Holm-correct the primacy and recency tests within
58 each comparison family. The fixed split contains 800 exploratory and 1,855 confirmatory questions.
59 Only the exploratory split is reported.

60 **Reproducibility and controls.** The harness vendors the official prompt builder and
61 best_subspan_em scorer at a pinned commit. It also records normalized exact match and first-
62 line extraction. Generation is greedy with temperature 0, at most 32 new tokens, fixed seeds, BF16
63 where supported, pinned model revisions, and one Mamba execution path. Each unique run pre-
64 serves prompts, raw generations, failures, token counts, software versions, and decoding settings.
65 The released key-value task provides an executed positive control: the pipeline detects a +0.38
66 Pythia start-versus-middle effect. Negative and distractor-order controls are implemented but have
67 no committed run artifacts, so we do not count them as passed controls.

68 4 Architecture comparisons

69 **Primary 2.8B comparison.** We compare pythia-2.8b (Biderman et al., 2023) with
70 mamba-2.8b and mamba2-2.7b. All are base models pretrained on the Pile (Gao et al., 2020),
71 have similar parameter counts, and use related tokenizers. Table 1 and Figure 1 show a positive
72 primacy edge for Pythia and no positive primacy edge for either Mamba model. The paired Mamba-
73 minus-Pythia primacy interactions are -0.053 and -0.070 (both Holm $p < 10^{-4}$), while Mamba-1
74 versus Mamba-2 is uncertain ($+0.017$, 95% CI $[-0.005, 0.038]$). All three models have positive
75 recency edges. These comparisons match corpus and approximate scale, but not depth, positional
76 encoding, state size, or checkpoint family.

Table 1: Pile-trained models on 800 exploratory questions. Intervals use a question-level paired bootstrap; p is Holm-corrected.

Model	Primacy (95% CI)	Recency (95% CI)	Floor / Ceiling
pythia-2.8b	+0.052 [0.032, 0.072], $p < 10^{-4}$	+0.075 [0.054, 0.097], $p < 10^{-4}$	0.091/0.640
mamba-2.8b	−0.001 [−0.016, 0.014], $p = 0.91$	+0.077 [0.058, 0.097], $p < 10^{-4}$	0.115/0.615
mamba2-2.7b	−0.018 [−0.034, −0.003], $p = 0.02$	+0.105 [0.086, 0.124], $p < 10^{-4}$	0.128/0.619

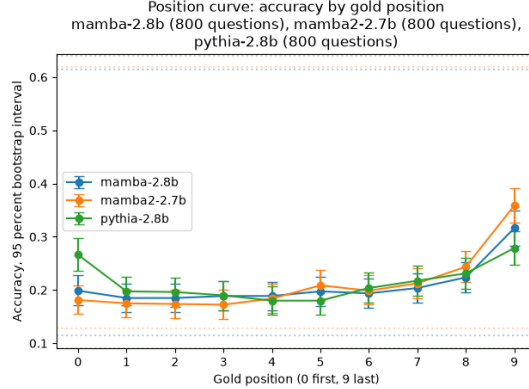


Figure 1: Accuracy versus gold position in the 2.8B comparison. Pythia rises at the start; all models rise at the end. Dotted lines are each model’s floor and ceiling.

77 **Matched 8B attention comparison.** Our strongest control uses NVIDIA’s pure and hybrid 8B
78 Mamba-2 checkpoints (Waleffe et al., 2024), which share training data, tokenizer, scale, depth, and
79 positional setup. The hybrid contains 7% attention layers. Within models, the pure model’s primacy
80 edge is uncertain (+0.008, $p = 0.36$) and the hybrid’s is positive (+0.027, $p = 0.007$). That
81 difference of significance is not itself a significant difference. The paired hybrid-minus-pure primacy
82 effect is +0.019 (95% CI [−0.006, 0.044], Holm $p = 0.144$), while the recency effect is −0.028
83 (95% CI [−0.055, −0.001], Holm $p = 0.089$). This direction supports the 2.8B observation but
84 does not isolate a causal attention effect.

Table 2: Matched 8B primacy comparison on 800 exploratory questions. The paired interaction is the relevant architecture test.

Contrast	Primacy	95% CI	Holm p
pure Mamba-2	+0.008	[−0.009, 0.024]	0.362
hybrid	+0.027	[0.008, 0.047]	0.007
hybrid minus pure	+0.019	[−0.006, 0.044]	0.144

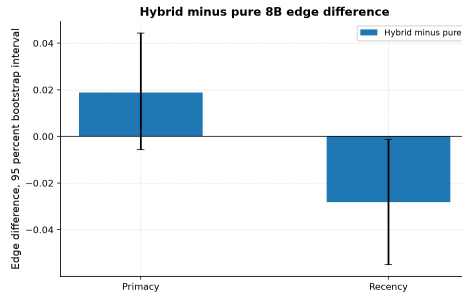


Figure 2: Hybrid-minus-pure 8B edge effects. The primacy interval crosses zero; recency decreases, but its Holm-corrected p -value is 0.089.

5 Scope checks

Scale. Across five approximate size pairs, the Pythia-minus-Mamba primacy gap is near zero at the two smallest scales and is +0.062, +0.069, and +0.053 in the three larger pairs (Figure 3). The smallest Pythia ceiling is only 0.009, so this is a descriptive capability-and-family trend, not evidence for a depth threshold.

Training corpus. Holding the Mamba-2.8B architecture fixed, the SlimPajama-minus-Pile primacy effect is -0.007 (95% CI $[-0.030, 0.016]$, $p = 0.568$) and the recency effect is -0.016 (95% CI $[-0.044, 0.012]$, Holm $p = 0.502$) (Soboleva et al., 2023). The floor differs substantially (0.270 versus 0.114), so this comparison finds no edge-shape change but does not rule out training-data effects.

Synthetic retrieval. On 50 RULER niah_single_1 examples at 2K tokens (Hsieh et al., 2024b), Pythia again has a primacy edge (+0.12, 95% CI $[0.05, 0.20]$). Both Mamba models score 1.0 at every depth, so this check reproduces the transformer pattern but cannot compare mixer curves on the synthetic task.

Descriptive system comparison. We also run Llama-3.1-8B (Dubey et al., 2024), Qwen2.5-7B (Yang et al., 2024), and Nemotron-H-8B (NVIDIA, 2025). All three show positive primacy edges (+0.076, +0.041, +0.067). Only Llama versus Qwen has a significant paired primacy interaction (+0.036, Holm $p = 0.016$). Because the systems differ on architecture, data, tokenizer, token count, depth, and positional encoding, these results establish prevalence, not an architectural explanation.

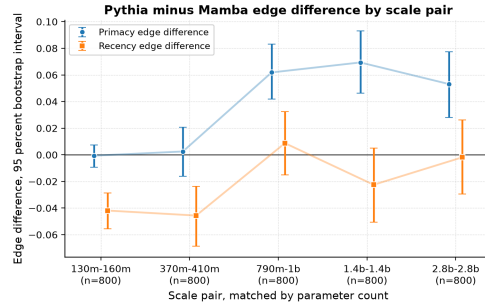


Figure 3: Pythia-minus-Mamba edge differences across approximate size pairs. The larger-model primacy gaps are descriptive because capability and several architectural factors vary together.

6 Mechanistic evidence and intervention

Attention sinks track primacy. Across the Pythia family, the 160M model has almost no token-0 attention sink (final-layer share 0.003), while the 2.8B model reaches 0.455 (Xiao et al., 2024). Pythia-410M has a small final-layer sink (0.034) and no significant primacy edge. Figure 4 is consistent with a sink mechanism, but the measurements share model scale as a confound and no valid sink ablation was completed.

Utilization, not storage, is a hypothesis. A balanced linear probe on frozen mid-depth states predicts edge versus middle position above its shuffled-label control for Pythia (0.603 versus 0.484) and Mamba (0.648 versus 0.508). Thus position information remains linearly decodable even when Mamba has no positive primacy edge. This supports a utilization hypothesis, but probe decodability does not prove that the model stores or uses the answer-bearing content.

Training-free prompt intervention. On 200 exploratory questions, repeating the question both before and after the documents increases Mamba primacy from 0.000 to +0.103 ($p < 10^{-4}$). Placing the question only before the documents does not create primacy (+0.020, $p = 0.366$), but reduces recency from +0.070 to +0.010. The intervention is training-free and practically useful, but it changes query repetition and token placement together, so it is not a clean test of fixed-state compression.

121 **Prompt sensitivity.** Changing the prompt template reduces Pythia primacy from +0.070 to
 122 +0.005 in the 200-question mechanism subset. This result warns against treating the measured
 123 magnitude as a template-invariant architectural constant.

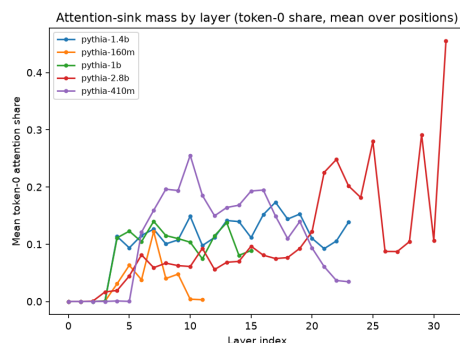


Figure 4: Mean token-0 attention share by Pythia layer and scale. Late-layer sink mass covaries with primacy but is not causally identified.

124 7 Discussion and limitations

125 The most secure result is the 2.8B exploratory architecture-position interaction. The matched 8B
 126 direction, sink correlation, and probe results jointly make attention-mediated utilization a plausible
 127 hypothesis, while prompt sensitivity shows that architecture is not the only determinant.

128 Several limitations set the confirmatory boundary. All analyses use the exploratory split, so the
 129 planned 1,855-question test must be run under a frozen analysis before confirmatory language is
 130 warranted. The 2.8B and scale comparisons retain depth, positional-encoding, state-size, and capa-
 131 bility confounds. The matched 8B paired contrast crosses zero after Holm correction, and its run
 132 manifest records a dirty source tree, weakening exact reproducibility. The negative and distractor-
 133 order controls lack committed outputs. A 50-generation blinded audit sample exists, but human
 134 labels were not completed. Finally, the attempted hybrid sink block did not alter the model because
 135 its custom attention path did not honor the hook; it is not an ablation result. The next decisive study
 136 should freeze exclusions and contrasts, complete these controls, repair the hybrid intervention, and
 137 evaluate the held-out split once.

138 8 Conclusion

139 On 800 exploratory Lost-in-the-Middle questions, Pythia-2.8B and two Pile-trained Mamba models
 140 have different primacy profiles under a paired ten-position evaluation. A matched 8B comparison is
 141 directionally consistent but statistically uncertain, so the present evidence supports an architecture-
 142 position interaction without establishing that attention causes it. The held-out split and completed
 143 control suite now provide a clear path from this exploratory workshop result to a confirmatory claim.

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